

FINAL MARK

**GIRRAWEEEN HIGH SCHOOL
MATHEMATICS
YEAR 12 HSC TASK 2 2017
ANSWERS COVER SHEET**

Name: _____

QUESTION	MARK	H2	H3	H4	H5	H6	H7	H8	H9
Q1 to Q5	/5				✓	✓	✓		✓
Q6	/13				✓				✓
Q7	/18				✓		✓		✓
Q8	/16				✓				✓
Q9	/10				✓				✓
Q10	/15				✓			✓	✓
Q11	/10				✓			✓	✓
Q12	/13				✓			✓	✓
TOTAL									
	/100				/100	/5	/23	/38	/100



GIRRAWEEEN HIGH SCHOOL

YEAR 12 – Task 2

2017

MATHEMATICS

Time allowed – 90 minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new page.

SECTION 1(Multiple Choice – 5 marks)

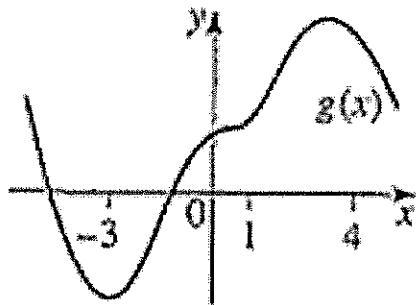
For questions 1 to 5, fill in the circle corresponding to the correct answer on your answer sheet.

1. If the graph of $g(x)$ has the following properties:

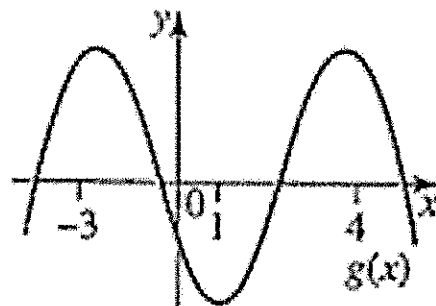
- (i) $g'(x) = 0$ if $x = -3, 1$ and 4
- (ii) $g'(x) < 0$ if $x < -3$ and $1 < x < 4$
- (iii) $g'(x) > 0$ for all other x

Then the graph of $g(x)$ could be

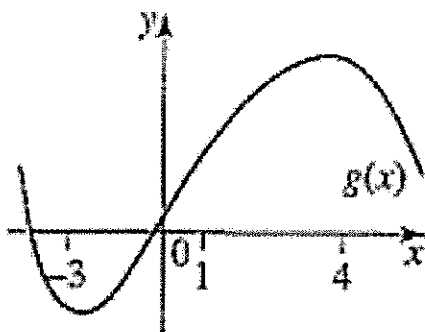
(A)



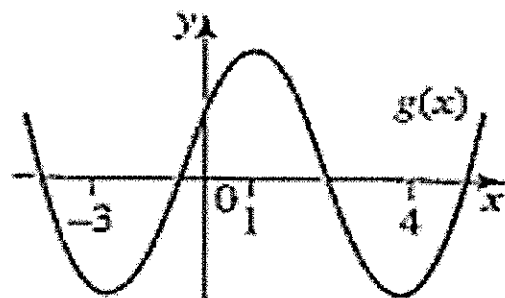
(C)



(B)



(D)



2. Evaluate $\int_1^5 (f(x) + 1)dx$, given that $\int_1^5 f(x)dx = 6$.

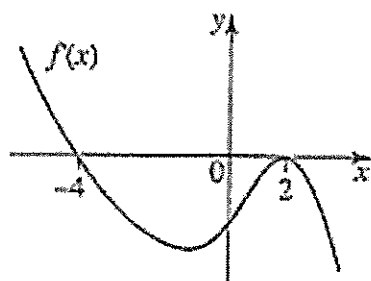
(A) 16

(B) 10

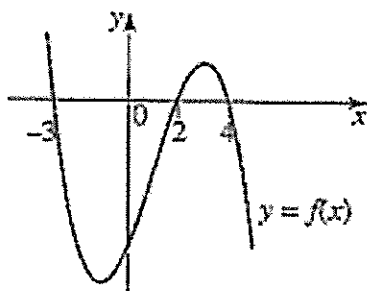
(C) 11

(D) 19

3. The graph of $f'(x)$ shown below indicates that the graph of $f(x)$ has



- (A) A turning point at $x = 2$ and $x = -4$.
 - (B) A turning point at $x = 2$ and point of inflexion at $x = -4$.
 - (C) A turning point at $x = -4$ and point of inflexion at $x = 2$.
 - (D) Two points of inflexion at $x = -4$ and $x = 2$.
4. The area between the curve, the x -axis and the lines $x = -3$ and $x = 4$ is equal to



- (A) $\int_{-3}^4 f(x) dx$
 - (B) $\int_{-3}^2 f(x) dx + \int_2^4 f(x) dx$
 - (C) $\int_2^4 f(x) dx + \int_2^{-3} f(x) dx$
 - (D) $\int_{-3}^2 f(x) dx - \int_2^4 f(x) dx$
5. An antiderivative or a primitive of $2(3x + 4)^{-4}$ is

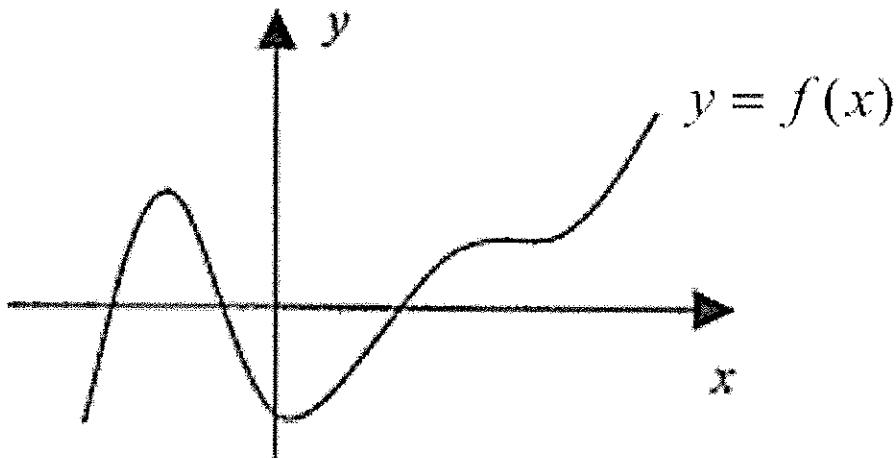
- (A) $-\frac{2}{3}(3x + 4)^{-3}$
- (B) $-\frac{2}{3}(3x + 4)^{-3} + 5$
- (C) $-\frac{2}{9}(3x + 4)^{-3}$
- (D) $-\frac{2}{9}(3x + 4)^{-5}$

Question 6 (13 marks)**Marks**

- (a) Determine the values of x for which $y = x^3 - 5x^2 + 3x + 2$ is decreasing. 3
- (b) Show that the curve $y = \frac{x}{x-1}$ has no stationary points. 2
- (c) Find $f'(x)$ and $f''(x)$ if $f(x) = \sqrt{1-3x}$ 4
- (d) For what values of x is the curve $f(x) = 2x^3 - 7x^2 - 5x + 4$ concave up? 4

Question 7 (18 marks)

- (a) Sketch the graph of $f'(x)$ (on separate sheet provided in the answer booklet) 4



- (b) For the curve $y = 6x^2 - 2x^3$, find:
- (i) the coordinates of the turning points and determine their nature. 4
- (ii) the coordinates of the point of inflexion. 3
- (iii) Sketch the curve showing the turning points, the point of inflexion and the points where the curve meets the x -axis. 5
- (iv) Determine the absolute maximum and minimum values of $f(x)$ for $-1 \leq x \leq 4$. 2

Question 8 (16 marks)

(a) The point $(2, -1)$ is a point of inflexion on the curve $y = x^3 + ax^2 + bx + 3$. Find

the values of a and b .

4

(b) Evaluate the following:

12

(i) $\int (x-2)(x-5) dx$

(ii) $\int \left(\frac{2}{x^4} - \frac{3}{x^3} \right) dx$

(iii) $\int_{-1}^4 (2x+3)^4 dx$

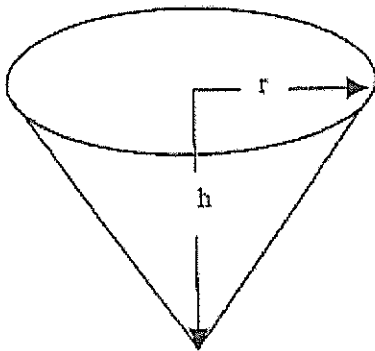
(iv) $\int_{-2}^1 \frac{4x^4 - x}{x} dx$

Question 9 (10 marks)

(a) The diagram represents a conical water Tower. The radius of the cone is r

and the height h . The volume of the cone is given by $V = \frac{1}{3}\pi r^2 h$. It is given that

$$2r + h = 60.$$



(i) Show that the volume (V) of the cone is $V = 20\pi r^2 - \frac{2}{3}\pi r^3$.

2

(ii) Find the maximum volume of the cone.

4

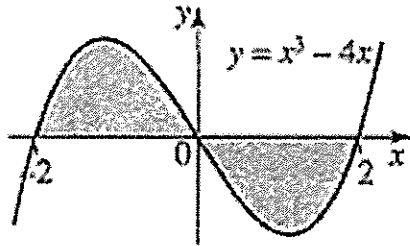
(b) At any point (x, y) on a curve, $\frac{d^2 y}{dx^2} = 12x + 6$. Find the equation of the curve if it

passes through the point $(-1, -2)$ and the gradient of the tangent at this point is 1. 4

Question 10 (15 marks)

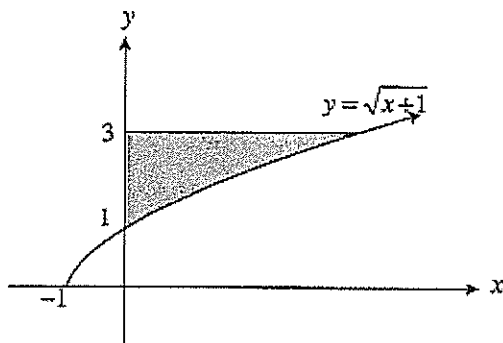
- (a) Find the area bounded by the curve $y = x^3 - 4x$ and the x -axis.

5

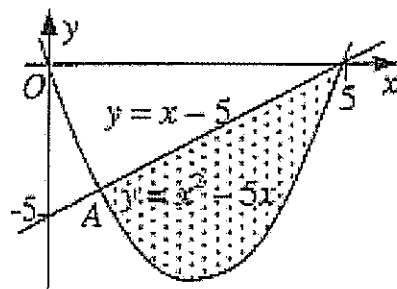


- (b) Find the area bounded by the curve $y = \sqrt{x+1}$, the y -axis and the lines $y = 1$ and $y = 3$.

4



- (c) The graphs of $y = x - 5$ and $y = x^2 - 5x$ intersect at the points $(5, 0)$ and A , as shown in the diagram.

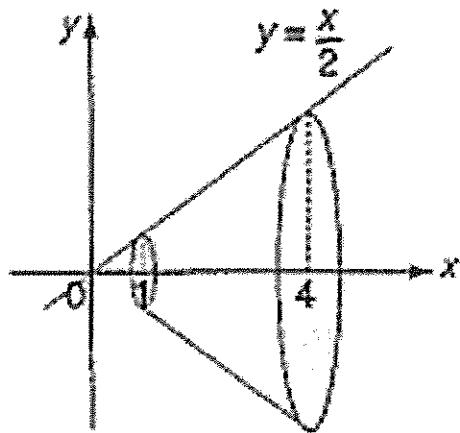


- (i) Find the coordinates of A . 2
- (ii) Find the area of the shaded region bounded by $y = x - 5$ and $y = x^2 - 5x$. 4

Question 11 (10 marks)

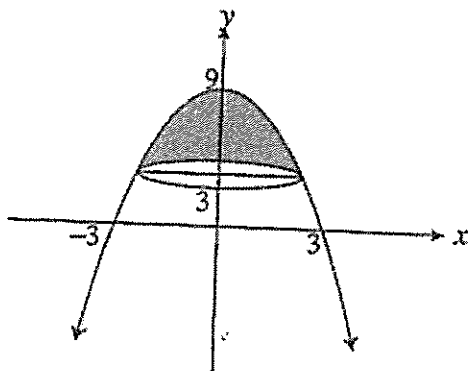
- (a) Find the volume of the solid generated when the section of the line $y = \frac{x}{2}$ between $x = 1$ and $x = 4$ is rotated about the x -axis.

3



- (b) Find the volume generated when the area bounded by the curve $y = 9 - x^2$ for $x \geq 0$, the y -axis and the line $y = 3$ is rotated about the y -axis.

3



- (c) (i) Differentiate $(x^4 - 1)^9$.

2

- (ii) Hence find $\int 2x^3 (x^4 - 1)^8 dx$.

2

Question 12 (13 marks)

(a) The table shows points on a continuous curve $y = f(x)$. Use the Trapezoidal Rule

to find the approximate value of $\int_3^4 f(x)dx$ correct to three decimal places. 3

x	3	3.2	3.4	3.6	3.8	4
y	7.19	7.62	8.41	8.74	9.26	9.78

(b) Evaluate $\int_0^2 \frac{1}{x+2} dx$ by Simpson's rule and by taking five function values.

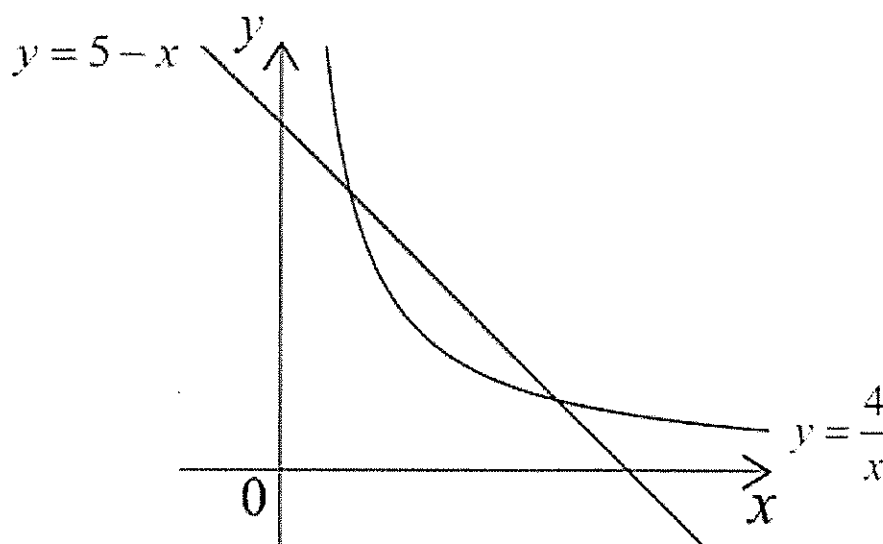
Write the answer correct to three decimal places.

5

(c) Find the volume of the solid formed when the area between $y = \frac{4}{x}$ and

$y = 5 - x$ is rotated about the x -axis.

5



END OF EXAMINATION

Year 12 Mathematics Test 2 2017 Solutions

Multiple Choice (5 marks)

1 D 2 B 3 C 4 C 5 C

Question 6 (13 marks)

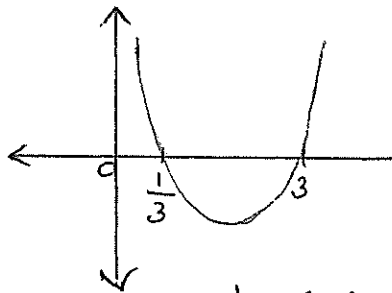
(a) $y = 2x^3 - 5x^2 + 3x + 2$

$$y' = 3x^2 - 10x + 3$$

For decreasing, $y' < 0$

$$3x^2 - 10x + 3 < 0$$

$$(3x-1)(x-3) < 0$$



(3)

$$\therefore \underline{\underline{\frac{1}{3} < x < 3}}$$

(b) $y = \frac{x}{x-1}$

$$y' = \frac{(x-1) \times 1 - x \times 1}{(x-1)^2}$$

$$= \frac{x-1-x}{(x-1)^2}$$

$$= \frac{-1}{(x-1)^2} \quad (2)$$

$$y' = 0 \Rightarrow \frac{-1}{(x-1)^2} = 0 \text{ which}$$

has no solutions.

$\therefore y = \frac{x}{x-1}$ has no stationary points.

(c) $f(x) = \sqrt{1-3x}$

$$f'(x) = \frac{1}{2\sqrt{1-3x}} x^{-3} = \frac{-3}{2\sqrt{1-3x}}$$

$$f''(x) = \frac{2\sqrt{1-3x} \times 0 - (-3) \times 2 \times \frac{1}{2} x^{-3}}{4(1-3x)}$$

$$= \frac{-9}{4(1-3x)\sqrt{1-3x}} = \frac{-9}{4(1-3x)^{\frac{3}{2}}}$$

$$= \frac{-9}{4\sqrt{(1-3x)^3}} \quad (4)$$

Alternative Method

$$f'(x) = \frac{-3}{2\sqrt{1-3x}}$$

$$= \frac{-3}{2} (1-3x)^{-\frac{1}{2}}$$

$$f''(x) = \frac{-3}{2} x^{-\frac{1}{2}} (1-3x)^{-\frac{3}{2}} x^{-3}$$

$$= \frac{-9x^{-1}}{4(1-3x)^{\frac{3}{2}}}$$

$$= \frac{-9}{4\sqrt{(1-3x)^3}}$$

$$(d) f(x) = 2x^3 - 7x^2 - 5x + 4$$

$$f'(x) = 6x^2 - 14x - 5$$

$$f''(x) = 12x - 14$$

For concave up, $f''(x) > 0$

$$12x - 14 > 0$$

$$12x > 14$$

$$x > \frac{14}{12}$$

(4)

$$\underline{\underline{x > \frac{7}{6}}}$$

Question 7 (18 marks)

(a) See last page

$$(b) y = 6x^2 - 2x^3$$

$$(i) y' = 12x - 6x^2$$

$$y' = 0 \Rightarrow 12x - 6x^2 = 0$$

$$6x(2-x) = 0$$

$$x = 0 \text{ or } 2-x = 2$$

When $x = 0$, $y = 0$

When $x = 2$, $y = 8$

Stationary points are

$(0,0)$ and $(2,8)$

$$y'' = 12 - 12x \quad \text{Page 2}$$

$$\text{when } x = 0, y'' = 12 > 0$$

$\therefore (0,0)$ is a minimum turning point

$$\text{when } x = 2, y'' = -12 < 0 \quad (4)$$

$\therefore (2,8)$ is a maximum turning point.

(ii) Possible points of inflexion are given by $y'' = 0$

$$12 - 12x = 0$$

$$x = 1$$

x	0	1	2
y''	12	0	-12

Concavity changes. (3)

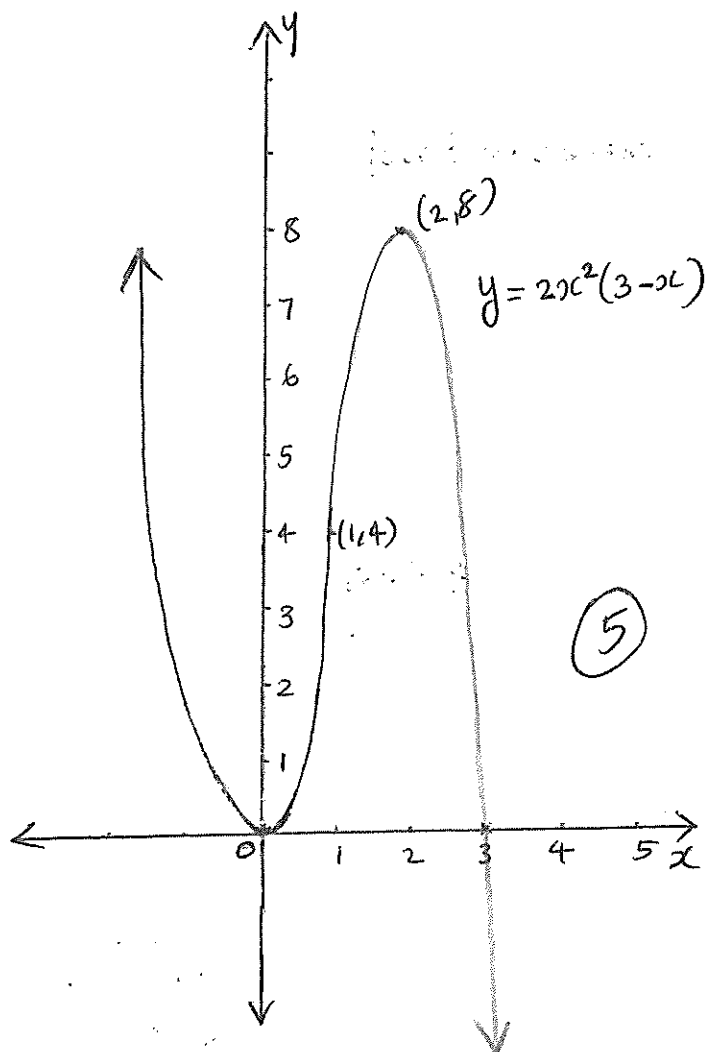
When $x = 1$, $y = 6 - 2 = 4$

$\therefore (1,4)$ is a point of inflexion.

(iii) x intercepts

$$6x^2 - 2x^3 = 0$$

$$2x^2(3-x) = 0 \quad x = 0, 3$$



(iv) when $x = -1$

$$y = 6 \times 1 - 2(-1) \\ = 8$$

when $x = 4$

$$y = 32 \times -1 = -32 \quad (2)$$

Absolute maximum is 8
and absolute minimum
is -32.

Question 8 (16 marks)

$$y = x^3 + ax^2 + bx + 3$$

$$y' = 3x^2 + 2ax + b$$

$$y'' = 6x + 2a$$

when $x = 2$, $y'' = 0$

$$0 = 12 + 2a$$

$$2a = -12$$

$$\therefore a = -6$$

$$y = x^3 - 6x^2 + bx + 3$$

when $x = 2$, $y = -1$

$$-1 = 8 - 24 + 2b + 3$$

$$2b = 12 \quad ; \quad b = 6$$

$$\underline{\underline{a = -6, b = 6}} \quad (4)$$

(b)(i) $\int (x-2)(x-5) dx$

$$= \int (x^2 - 7x + 10) dx$$

$$= \underline{\underline{\frac{x^3}{3} - \frac{7x^2}{2} + 10x + C}} \quad (3)$$

(ii) $\int \left(\frac{2}{x^4} - \frac{3}{x^3} \right) dx$

$$= \int (2x^{-4} - 3x^{-3}) dx$$

$$= 2 \frac{x^{-3}}{-3} - 3 \times \frac{x^{-2}}{-2} \quad (3)$$

$$= \underline{\underline{\frac{-2}{3x^3} + \frac{3}{2x^2} + C}}$$

$$(iii) \int_{-1}^4 (2x+3)^4 dx$$

$$= \left[\frac{(2x+3)^5}{10} \right]_{-1}^4$$

$$= \frac{1}{10} \left[(2x+3)^5 \right]_{-1}^4$$

$$= \frac{1}{10} (11^5 - 1) \quad (3)$$

$$= 16105$$

$$(iv) \int_{-2}^1 \frac{4x^4 - x}{x} dx$$

$$= \int_{-2}^1 (4x^3 - 1) dx$$

$$= \left[\frac{4x^4}{4} - x \right]_{-2}^1$$

$$= [x^4 - x]_{-2}^1 \quad (3)$$

$$= (1-1) - (16+2)$$

$$= -18$$

Question 9 (10 marks)

$$(i) 2r + h = 60$$

$$h = 60 - 2r$$

$$V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi r^2 (60 - 2r)$$

page 4

$$= \frac{60\pi r^2}{3} - \frac{2}{3} \pi r^3$$

$$= 20\pi r^2 - \frac{2}{3} \pi r^3 \quad (2)$$

$$(ii) \frac{dV}{dr} = 40\pi r - 2\pi r^2$$

$$= 2\pi r(20 - r)$$

$$= 0 \text{ when } r = 0 \text{ or } r = 20$$

$$\frac{d^2V}{dr^2} = 40\pi - 4\pi r$$

$$\text{when } r = 20, \frac{d^2V}{dr^2} = 40\pi - 4\pi \times 20$$

$$= 40\pi - 80\pi$$

$$= -40\pi < 0$$

$\therefore$ Volume is maximum when $r = 20$

Maximum volume

$$= \frac{1}{3} \pi (20)^2 \times 20 \quad (4)$$

$$= \frac{8000\pi}{3} \text{ cubic units.}$$

$$(b) \frac{d^2y}{dx^2} = 12x + 6$$

$$\frac{dy}{dx} = 12 \frac{x^2}{2} + 6x + C$$

$$= 6x^2 + 6x + C$$

$$\text{when } x = -1, y' = 1$$

$$1 = 6 - 6 + C \quad \therefore C = 1$$

$$\frac{dy}{dx} = 6x^2 + 6x + 1$$

$$y = \frac{6x^3}{3} + \frac{6x^2}{2} + x + D$$

$$= 2x^3 + 3x^2 + x + D$$

When $x = -1$, $y = -2$

$$-2 = -2 + 3 - 1 + D$$

$$D = -2$$

(4)

$$\therefore y = 2x^3 + 3x^2 + x - 2$$

Question 10 (15 marks)

Shaded area

$$= \int_{-2}^0 (x^3 - 4x) dx - \int_0^2 (x^3 - 4x) dx$$

I_1

I_2

$$= (9 - 3) - \frac{1}{3} + 1 = 6 \frac{2}{3} \text{ square units}$$

$$I_1 = \left[\frac{x^4}{4} - \frac{4x^2}{2} \right]_{-2}^0$$

$$= \left[\frac{x^4}{4} - 2x^2 \right]_{-2}^0$$

$$0 - \left(\frac{16}{4} - 2 \times 4 \right)$$

$$= 4$$

$$I_2 = \left[\frac{x^4}{4} - 2x^2 \right]_0^2$$

$$= \left(\frac{16}{4} - 8 \right) - 0$$

$$= -4$$

(5)

$$I_1 - I_2 = 4 - (-4)$$

$$= 8 \text{ square units}$$

(b) $y = \sqrt{x+1}$

$$y^2 = x+1 \quad ; \quad x = y^2 - 1$$

$$\text{Required area} = \int_1^3 x dy$$

$$= \int_1^3 (y^2 - 1) dy$$

$$= \left[\frac{y^3}{3} - y \right]_1^3$$

$$= \left(\frac{3^3}{3} - 3 \right) - \left(\frac{1}{3} - 1 \right)$$

(4)

(c) (i) $x^2 - 5x = x - 5$

$$x^2 - 6x + 5 = 0$$

$$(x-1)(x-5) = 0$$

$$x = 1 \quad \text{or} \quad x = 5$$

When $x = 1$, $y = 1 - 5 = -4$

$$\therefore A(1, -4)$$

(2)

(ii) Shaded area

$$= \int_1^5 (x - 5 - x^2 + 5x) dx$$

$$= \int_1^5 (-x^2 + 6x - 5) dx$$

$$= \left[-\frac{x^3}{3} + 3x^2 - 5x \right]_1^5$$

(4)

$$= \left(-\frac{125}{3} + 75 - 25 \right) - \left(-\frac{1}{3} + 3 - 5 \right)$$

$$= \frac{32}{3} \text{ square units}$$

Question 11 (10 marks)

$$\begin{aligned}
 (a) \quad V &= \pi \int_1^4 \frac{x^2}{4} dx \\
 &= \frac{\pi}{4} \left[\frac{x^3}{3} \right]_1^4 \\
 &= \frac{\pi}{12} [x^3]_1^4 \\
 &= \frac{\pi}{12} (64-1) \quad (3) \\
 &= \frac{21\pi}{4} \text{ cubic units}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad y &= 9-x^2 \\
 x^2 &= 9-y \\
 V &= \pi \int_3^9 (9-y) dy \quad (3) \\
 &= \pi \left[9y - \frac{y^2}{2} \right]_3^9 \\
 &= \pi \left[\left(81 - \frac{81}{2} \right) - \left(27 - \frac{9}{2} \right) \right] \\
 &= 18\pi \text{ cubic units}
 \end{aligned}$$

Question 12 (13 marks) page 6

$$\begin{aligned}
 (a) \quad &\int_0^3 f(x) dx \\
 &= \frac{0.2}{2} \left\{ 7.19 + 9.78 + 2(7.62 + 8.41 + 8.74 + 9.26) \right\} \\
 &= \underline{\underline{8.503}} \quad (3)
 \end{aligned}$$

$$(b) \quad n=4 \quad h = \frac{2-0}{4} = 0.5$$

$$\begin{array}{cccccc}
 0 & \frac{1}{2} & 1 & \frac{3}{2} & 2 \\
 \frac{1}{2} & \frac{2}{5} & \frac{1}{3} & \frac{2}{7} & \frac{1}{4}
 \end{array}$$

$$\begin{aligned}
 &\int_0^2 \frac{1}{x+2} dx \\
 &= \frac{0.5}{3} \left\{ \left(\frac{1}{2} + \frac{1}{4} \right) + 2 \left(\frac{1}{3} \right) + 4 \left(\frac{2}{5} + \frac{2}{7} \right) \right\} \\
 &= \underline{\underline{0.693}} \quad (5)
 \end{aligned}$$

$$(i) \frac{d}{dx} (x^4-1)^9 = 9(x^4-1)^8 \times 4x^3 = \underline{\underline{36x^3(x^4-1)^8}} \quad (2)$$

$$(ii) \int 36x^3 (x^4-1)^8 dx = (x^4-1)^9 + C$$

$$18 \int 2x^3 (x^4-1)^8 dx = (x^4-1)^9 + C$$

$$\int 2x^3 (x^4-1)^8 dx = \underline{\underline{\frac{(x^4-1)^9}{18}}} + C \quad (2)$$

(1) Points of intersection

$$\frac{4}{x} = 5 - x$$

$$4 = x(5 - x)$$

$$4 = 5x - x^2$$

$$x^2 - 5x + 4 = 0$$

$$(x - 4)(x - 1) = 0$$

$$x = 4 \text{ or } x = 1$$

$$V = \pi \int_1^4 (5 - x)^2 dx - \pi \int_1^4 \left(\frac{4}{x}\right)^2 dx$$

$$= \pi \int_1^4 \left(25 - 10x + x^2 - \frac{16}{x^2}\right) dx \quad (5)$$

$$= \pi \left[25x - 5x^2 + \frac{x^3}{3} + \frac{16}{x} \right]_1^4$$

$$= \pi \left\{ \left(100 - 80 + \frac{64}{3} + 4\right) - \left(25 - 5 + \frac{1}{3} + 16\right) \right\}$$

$$= \underline{\underline{9\pi \text{ cubic units}}}$$

Question 7(a)

